

Requirements for Providers and Deployers of High-Risk AI Systems under the EU AI Act

The EU Artificial Intelligence Act establishes a comprehensive set of obligations for **high-risk AI systems**. These requirements are designed to ensure that such systems remain safe, reliable, transparent and subject to effective human control throughout their lifecycle.

The regulation distinguishes between two important actors. The **provider** is generally the organisation that develops an AI system, or has it developed, and places it on the market or puts it into service under its own name or trademark. The **deployer** is the organisation or person using the AI system under its authority in a professional context.

Providers carry most of the obligations relating to the design, testing and conformity of high-risk systems. Deployers are mainly responsible for ensuring that these systems are used correctly, monitored appropriately and operated with adequate human oversight.

1. Risk Management – Article 9

Under **Article 9**, every high-risk AI system must have a documented **risk management system**.

Risk management is not a one-time assessment performed before launching an AI product. It must be a **continuous and iterative process covering the entire lifecycle of the system**.

Providers must identify and analyse known and reasonably foreseeable risks to health, safety and fundamental rights. They must evaluate risks arising both from the intended use of the system and from reasonably foreseeable misuse.

Appropriate measures must then be implemented to eliminate or reduce these risks as far as possible. The provider must also test the system against predefined metrics and thresholds and reconsider risks when new information becomes available, particularly through post-market monitoring.

For example, a provider developing an AI recruitment system should examine whether the system could systematically disadvantage certain categories of candidates. The provider should identify this risk, test the system, implement mitigation measures and continue monitoring its performance after deployment.

Special consideration must also be given to potential adverse impacts on children and other vulnerable groups.

2. Data and Data Governance – Article 10

The quality of an AI system depends heavily on the quality of the data used to develop it. **Article 10** therefore establishes requirements concerning training, validation and testing datasets.

For high-risk systems using models trained on data, datasets must be subject to appropriate **data governance and management practices**.

Providers must consider issues such as data collection processes, the origin of the data, preparation and processing operations, assumptions concerning what the data represents, possible biases and measures adopted to detect and mitigate those biases.

Training, validation and testing datasets must be relevant, sufficiently representative and, to the best extent possible, free of errors and complete in view of the intended purpose of the system.

The data should also reflect the geographical, behavioural or contextual characteristics of the population in which the AI system is intended to operate when those characteristics are relevant.

For example, an AI system designed to evaluate European students should not automatically be considered reliable simply because it performed well on a dataset composed of students from a very different educational and cultural environment.

Data governance is therefore closely connected with both **AI performance and protection against discrimination**.

3. Technical Documentation – Article 11

Under **Article 11**, providers must prepare detailed **technical documentation** before a high-risk AI system is placed on the market or put into service.

The documentation must demonstrate that the system complies with the requirements of the AI Act and must provide authorities with the information necessary to assess that compliance.

It should describe areas such as the system's intended purpose, its design and development, the methods and technologies used, its architecture, data requirements, performance metrics, testing and validation procedures, risk-management measures and monitoring arrangements.

The documentation must also be maintained and updated when necessary.

This requirement creates an important principle of accountability: a provider must not simply claim that an AI system is safe. It must be capable of **documenting how the system was designed, tested and validated and how its risks are controlled**.

4. Record-Keeping – Article 12

Article 12 requires high-risk AI systems to technically allow for the **automatic recording of events**, generally referred to as logs, throughout the lifetime of the system.

These logs must enable an appropriate level of traceability.

Record-keeping can help identify situations in which the system has operated outside its intended conditions, facilitate post-market monitoring and assist in investigating incidents or unexpected behaviour.

For example, if an AI system used in recruitment produces a controversial recommendation, appropriate logs can help determine what occurred when the system processed the application.

Logs therefore provide a form of **audit trail**, helping providers, deployers and supervisory authorities understand how a system behaved.

5. Transparency and Information to Deployers – Article 13

Under **Article 13**, high-risk AI systems must be sufficiently transparent to allow deployers to **interpret their outputs and use them appropriately**.

Providers must supply clear and understandable instructions for use.

These instructions must include relevant information about the system's intended purpose, capabilities and limitations, expected levels of accuracy, robustness and cybersecurity, known risks, input-data requirements and circumstances that may affect performance.

Providers must also explain the human oversight measures that should be implemented and provide relevant information enabling deployers to interpret the system's output.

This does not necessarily mean that every technical aspect of a complex AI model must be understandable to every user. Rather, deployers must receive enough information to understand what the system can and cannot reliably do and how it should be used.

Transparency therefore creates an essential connection between the responsibilities of the **provider and the deployer**.

6. Human Oversight – Article 14

One of the central principles of the AI Act is that important decisions involving high-risk AI should remain subject to meaningful **human oversight**.

Under **Article 14**, high-risk systems must be designed so that they can be effectively overseen by natural persons while they are being used.

People responsible for oversight should be able to understand the system's capabilities and limitations, monitor its operation and recognise possible anomalies or unexpected behaviour.

They must also be able, where appropriate, to **disregard, override or reverse the system's output**, or interrupt the operation of the system.

A particularly important concept is the danger of **automation bias**: humans may have a tendency to trust an automated recommendation simply because it was produced by an apparently sophisticated AI system.

For example, a recruitment manager should not automatically reject a candidate simply because an AI tool gives the application a low score. Effective human oversight requires the organisation to retain the ability to critically assess the AI recommendation.

The appropriate level of oversight depends on the risks, autonomy and context of use of the system.

7. Accuracy, Robustness and Cybersecurity – Article 15

Under **Article 15**, high-risk AI systems must achieve an appropriate level of **accuracy, robustness and cybersecurity** and maintain appropriate performance throughout their lifecycle.

Providers must define relevant accuracy metrics and communicate the expected levels of accuracy in the instructions supplied with the system.

Robustness means that the AI system should continue to perform reliably when confronted with errors, faults, inconsistencies or unexpected situations.

Cybersecurity is particularly important because AI systems can be exposed to attacks that exploit their specific vulnerabilities. These may include attempts to manipulate training data, inputs or model behaviour.

Providers must therefore design high-risk AI systems with appropriate technical and organisational measures to protect them against malicious interference.

These three concepts are closely related: a high-risk system must not only perform accurately under ideal laboratory conditions; it must remain sufficiently reliable and secure in the real environment in which it is deployed.

8. Responsibilities of Deployers

Although many of the technical requirements fall primarily on providers, deployers also have significant responsibilities.

Deployers must generally follow the provider's **instructions for use**, implement the required human oversight and assign oversight to individuals with appropriate competence, training and authority.

They must ensure that relevant input data under their control is appropriate to the system's intended purpose and monitor the operation of the system on the basis of the instructions supplied by the provider.

Where there is reason to believe that the use of the system may create a risk, appropriate action must be taken. Depending on the circumstances, this may include suspending use and informing the provider or other relevant parties.

Certain deployers may also have additional obligations concerning record keeping, informing workers that they are subject to a high-risk AI system, or carrying out a **fundamental rights impact assessment** before deploying certain systems.

The AI Act therefore establishes a chain of responsibility: **providers are responsible for building compliant systems, while deployers are responsible for using them appropriately.**

9. Conformity Assessment

Before a high-risk AI system is placed on the EU market or put into service, the provider must ensure that it undergoes the appropriate **conformity assessment procedure**, as provided principally by **Article 43**.

The purpose of conformity assessment is to verify that the AI system complies with the mandatory requirements of the AI Act, including those concerning risk management, data governance, documentation, record-keeping, transparency, human oversight, accuracy, robustness and cybersecurity.

Depending on the type of high-risk system and the applicable standards, conformity may in some situations be assessed through an **internal control procedure**. In other circumstances, particularly for certain biometric systems or when the relevant conditions for internal assessment are not satisfied, assessment of the quality-management system and technical documentation requires the involvement of an independent **notified body**.

High-risk AI systems that are safety components of products already regulated by EU product-safety legislation are generally assessed within the conformity-assessment procedures applicable to those products.

Once conformity has been demonstrated, the provider must draw up an **EU Declaration of Conformity** and apply the **CE marking** where required. The system must also be registered when the relevant registration obligations apply.

Conformity assessment is not necessarily the end of the process. If a high-risk AI system undergoes a **substantial modification**, a new conformity assessment may be required.

This reflects a central principle of the AI Act: compliance is not simply a certificate obtained once before launch. It is an ongoing responsibility throughout the lifecycle of the AI system.

Conclusion

The requirements imposed on high-risk AI systems create a comprehensive framework covering their entire lifecycle.

Articles 9 to 15 address the core technical and organisational requirements: risk management, data governance, technical documentation, logging, transparency, human oversight, accuracy, robustness and cybersecurity. Providers must incorporate these requirements into the design and development of their systems and demonstrate compliance through the relevant conformity assessment procedure.

Deployers, in turn, must use high-risk AI systems according to their instructions, provide appropriate human oversight and monitor their operation.

The overall logic can therefore be summarised as a continuous compliance cycle:

identify risks → control data → document the system → maintain records → inform deployers → ensure human oversight → test accuracy, robustness and cybersecurity → demonstrate conformity → monitor the system after deployment.

This lifecycle approach is one of the defining characteristics of the EU AI Act. For high-risk AI, compliance is not limited to the technology itself: it concerns the entire process through which the system is designed, tested, deployed, supervised and maintained.